MIL-F-5509D
AMENDMENT 4
30 NOVEMBER 1987
SUPERSEDING
AMENDMENT 3
15 February 1984

MILITARY SPECIFICATION

FITTINGS, FLARED TUBE, FLUID CONNECTION

This amendment forms a part of Military Specification MIL-F-5509D, dated 16 June 1978, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

2.1, delete first sentence and substitute the following 2.1

"2.1 Government documents."

Add the following new paragraph 2.1.1:

"2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation."

Add to bottom of page:

"Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: San Antonio ALC/MMEDO, Kelly AFB, TX 78241 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter."

PAGE 2

Delete "QQ-S-637, Steel Bar Carbon, Cold Finished (Standard Quality, Free Machining)."

Delete title opposite PPP-B-566 and substitute "Box, Folding, Paperboard."

Delete title opposite PPP-B-585 and substitute "Box, Wood, Wirebound."

Delete title opposite PPP-B-621 and substitute "Box, Wood, Nailed and Lock-corner."

Delete title opposite PPP-B-636 and substitute "Box, Shipping Fiberboard."

Delete title opposite PPP-B-676 and substitute "Boxes, Setup."

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Add the following:

"QQ-S-781 Strapping, Steel, and Seals."

Delete title opposite MIL-P-116 and substitute "Preservation, Methods of."

PAGE 3

Under SPECIFICATION, MILITARY, delete "MIL-B-13239, Barrier Material, Waterproofed, Flexible, All Temperatures, Heat Sealable."

*Add: "MIL-B-131, Barrier, Materials Watervaporproof Flexible, Heat Sealable."

"MIL-H-46170, Hydraulic Fluid, Rust Inhibited, Fire-Resistant, Synthetic Hydrocarbon."

Under Standards Military add:

"MIL-STD-147 Palletized Unit Loads
MIL-STD-794 Parts and Equipment, Procedures for Packaging and Packing of."

PAGE 4

Delete parenthetical source paragraph in its entirety and substitute the following:

"(Copies of specifications, standards, handbooks, drawings, publications and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)"

2.2 Delete lines 1 and 2 and substitute the following:

"2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the non-government documents which is current on the date of the solicitation."

Under paragraph 2.2, add: "AMERICAN SOCIETY FOR TESTING AND MATERIAL (ASTM)"

"ASTM A-108 Steel Bars, Carbon, Cold Finished, Standard Quality"

"ASTM D-3951 Packaging, Commercial"

"(Copies of the above may be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)"

Under SOCIETY OF AUTOMOTIVE ENGINEERS, INC., delete the title of AMS 4928 and substitute "Bars, Titanium Alloy, Forgings and Rings 6AL-4V Annealed 120,000 PSI (825 Mpa) Yield Strength."

Under SOCIETY OF AUTOMOTIVE ENGINEERS, INC., add:

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"AS 1376 Wrench Pads for Fluid Fittings Alternate Machined Shapes of, Design Std for."

*Add the following new paragraph 2.3 as follows:

"2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained."

PAGE 5

3.1, change to read: "Material. Fittings shall be fabricated of materials listed in Table I or as specified on the applicable drawings."

Paragraph 3.1.2.1, delete in its entirety and substitute the following:

"3.1.2.1 Aluminum alloy. Aluminum alloy fittings and nuts shall be supplied in the final temper shown in Table I. Process Control shall be in accordance with MIL-H-6088."

PAGE 6

Delete page 6 in its entirety and substitute the attached page 6. Page 6 forms page 4 (from Table I to the end of the page) of this amendment.

PAGE 7

Delete "3.2.3 Threads." and substitute the following:

"3.2.3 Threads. Pipe threads shall conform to MIL-P-7105. Straight threads shall conform to MIL-S-8879 except root radius is not required on incomplete threads. Threads on carbon steel, CRES and titanium alloy may be cut, ground or rolled. Threads on aluminum alloy parts shall be cut or ground. Threads on brass or bronze manganese shall be cut or ground."

Paragraph 3.2.3.1, line 5, change "III" to "II."

Paragraph 3.3.1, delete and substitute the following:

"3.3.1 Aluminum alloy fittings. Aluminum alloy fittings, sleeves, and nuts shall be anodized in accordance with MIL-A-8625, Type II, Class 2 and be duplex sealed."

PAGE 8

Change Table "III" to "II."

3.3.2, change to read: "Carbon steel fittings. Carbon steel fittings, sleeves and nuts shall be cadmium plated in accordance with QQ-P-416, Type II, Class 2. All such carbon steel fittings, sleeves and nuts shall be dipped in oil conforming to MIL-H-6083 or MIL-H-46170. Fluid passage holes, other openings and internal threads shall not be subject to a plating thickness

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requirement and may exhibit bare areas provided that they are protected with a light film of oil."

TABLE I. Materials.

Material	Type of Part	Form	Specification	Material Code
Aluminum Alloy	Straights 4/ and Shapes	Bars and rods	QQ-A-225/4 (2014) Temper T6	D
			QQ-A-225/6 (2024) Temper T6 or T851	D
			QQ-A-225/9 (7075) Temper T73	W
		Forgings	QQ-A-367 (2014) Temper T6	D
			QQ-A-367 (7075) Temper T73	W
Copper-base	Straights and Shapes	Bars, rods, forgings	1/QQ-B-728 Half hard	B or none
	Straights	Bars and rods	QQ-B-626 Alloy 360 Half hard	
	Shapes	Forgings	QQ-B-626 Alloy 377	
Steel	Straights	Tubing seamless	MIL-T-5066 (1025)	None
	Straights and shapes	Bars, rods, forgings	ASTM A108 (1137) ASTM A108 (1141) MIL-S-6758 (4130) MIL-S-5626 (4140)	None
Corrosion-resisting and heat-resisting steels	Straights and shapes	Bars, rods, forgings	2/QQ-S-763 Class 304	J
			2/QQ-S-763 Class 316	K
			2/QQ-S-763 Class 347	S
Titanium alloy	Straights and shapes	Bars, rods, forgings	3/AMS 4928 (6Al4V annealed)	T

1/ With Bronze forgings, a soft temper will be permitted in the finished part.
2/ These alloys shall conform to FED-STD-66.
3/ Titanium hex bars shall be cold finished with tolerances per AMS 2241.
4/ Wrench pad dimensions of shapes machined from bar stock shall conform with AS 1376.

3.2 Design and dimensions. The design and dimensions of the fittings shall be in accordance with the applicable drawings. The bosses and tube and hose ends of fittings shall be in accordance with MS24385, MS24386, MS33649, MS33656, MS33657, MS33658 or MS33677, as applicable. Dimensional requirements are applicable after heat treatment and protective finishing.

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PAGE 9

Under paragraph 3.4.5 Color identification., change line 4 to read, "Aluminum alloy 2014 and 2024 - Blue 1/." Line 5, add "1/" after Brown. Bottom of paragraph add, "1/ Duplex seal may distort color of aluminum anodize."

PAGE 10

4.1, delete in its entirety and add the following new paragraph 4.1 and 4.1.1.

"4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements."

"4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling for quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material."

PAGE 11

4.4.4, line 5, change "IV" to "III."

4.4.5, line 3, delete "P-C-43" and substitute "P-C-437."

PAGE 12

Change Table "IV" to "III."

4.6, line 5, change "V" to "IV."

Change Table "V" to "IV."

PAGE 14

*5.1 Delete and substitute the following:

"5.1 Preservation. Preservation shall be level A, C or Commercial. (see 6.2)."

5.1.1.2, line 4, change "Table VI" to read "Table V."

Change "Table VI" to read "V."

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5.1.1.3, line 5, change "Table VII" to read "Table VI."

PAGE 15

Change "Table VII, Quantity per Unit Container" to read "Table VI, Quantity per Unit Container."

*5.1.2 delete in its entirety and substitute the following:

"5.1.2 Commercial. The commercial preservation of the fittings shall be in accordance with requirements of ASTM-D-3951."

*Delete paragraphs 5.2, 5.2.1 and 5.2.2 in their entirely and substitute the following paragraphs 5.2, 5.2.1 and 5.2.2:

* "5.2 Packing. Packing shall be Level A, B, C or Commercial as specified (see 6.2).

* 5.2.1 Level A. Fittings packaged as specified herein shall be packed in overseas-type shipping containers conforming to PPP-B-585, PPP-B-591, PPP-B-601 or PPP-B-621. Shipping containers, insofar as possible, shall contain the same number of articles, shall be uniform in shape and size and snugly packed, shall be of minimum cube and tare consistent with the protection required, and shall contain identical quantities. The gross weight of the fully packed exterior shipping container shall not exceed approximately 200 pounds, except fiberboard containers, which shall not exceed the weight limitations of the applicable container specification. Plywood, when used, shall be Type I or II, Class 2, of NN-P-530. Strapping and closures shall be in accordance with the appendix to the applicable container specification. Containers conforming to PPP-B-585 or PPP-B-621 shall be provided with a case liner conforming to MIL-L-10547, except that material shall conform to MIL-B-131 and shall be sealed in accordance with the appendix to MIL-L-10547.

*5.2.2 Level B. Fittings packaged as specified herein shall be packed in a domestic-type shipping container conforming to PPP-B-585, PPP-B-591, PPP-B-601 or PPP-B-621. Closures shall be in accordance with the appendix to the applicable container specification. The gross weight of the shipping container, when packed for shipment, shall not exceed 200 pounds."

PAGE 16

*5.2.3 delete in its entirety and substitute the following:

"5.2.3 Level C. Fittings packaged as specified herein shall be packed in domestic-type shipping containers conforming to PPP-B-585, PPP-B-591, PPP-B-601, PPP-B-621, or PPP-B-636. Closures shall be in accordance with the appendix to the applicable container specification. Fiberboard, when used, shall have a minimum shipping container, when packed for shipment, shall not exceed 200 pounds; fiberboards containers shall not exceed the weight limitations specified in the applicable container specification."

* Add the following new paragraphs 5.2.4 and 5.2.5.

*"5.2.4 Commercial. The preserved fittings shall be packed in accordance with the requirements of ASTM-D-3951."

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* "5.2.5 Palletized loads. When specified (see 6.2). Palletized loads, commensurate with the level of packing specified in the contract or order, shall be palletized in accordance with MIL-STD-147. Palletized loads shall be uniform in size and quantities to the greatest extent possible. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer."

* 5.3. delete and substitute the following:

* "5.3 Marking. In addition to any special or other identification marking required by the contract (see 6.2), each unit pack, intermediate and exterior container and unitized load shall be marked in accordance with MIL-STD-129 or ASTM-D-3951 as applicable."

*Add the following new paragraph 5.4, 5.4.1 and 5.4.2:

* 5.4 General.

5.4.1 Exterior containers. Exterior containers (see 5.2.1, 5.2.2, 5.2.3 and 5.2.4) shall be of minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

* 5.4.2 Packaging Inspection. The inspection of these packaging requirements shall be in accordance with 4.6."

* 6.2d delete in its entirety and substitute the following: "d. Selection of applicable levels of preservation, packaging, packing, marking and commercial/industrial (see 5.1, 5.2 and 5.3)."

*. Under 6.2 add the following new subparagraph e: "e. When palletized loads are required (see 5.4)."

PAGE 17

6.3 delete in its entirety. Add the following new paragraph 6.3.

"6.3 Subject term (key word) listing.

Fittings
Fittings, flared tube
Fittings, flared tube, fluid connection."

Last line delete "International Interest: (see 6.3)."

The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccurancies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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